

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

MSE 3201
(Materials Characterization)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) How to determine the depth profile of any metallic sample using XPS? Explain with necessary figure. (10)
(b) Explain the importance of analyzing crystal in wavelength dispersive spectroscopy (WDS). (10)
(c) Mention the problems of analyzing electrically insulating material using AES or XPS. How to overcome this problem? (15)
2. (a) What is the effect of tip to sample separation of STM? (08)
(b) Mention the tip preparation methods for SPM. Write down the differences between constant current mode and constant height mode. (15)
(c) Describe the DLTS measurement theory for electrical characterization with appropriate diagram. (12)
3. (a) What are the selection rules for characteristic X-ray emission? (08)
(b) "There is an intrinsic disadvantage for detection of light elements using X-ray spectroscopy"- explain this using the concept of fluorescent yield. (12)
(c) "Sample preparation and working atmosphere play the major role in EDS" - justify the statement. (15)
4. (a) Explain the effects of scanning rate in DSC using appropriate diagram. (15)
(b) Is it possible to construct a phase diagram using DSC or DTA? If so, justify your answer. (10)
(c) Mention the measurement techniques for bulk resistivity. What are the problems with DLTS in electrical characterization? (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) What is structure factor. Calculate the structure factor for a FCC unit cell and show five planes from where reflection may occur. (15)
- (b) Derive an equation that is used to determine the particle size of crystalline materials. (12)
- (c) Draw an XRD curve for crystalline solid and amorphous solid. (08)
6. (a) Write down the steps of specimen preparation for carbon replica method with necessary diagram. (15)
- (b) How to prepare a ceramic sample for TEM study? (10)
- (c) Explain the diffraction contrast that arise in TEM image. (10)
7. (a) What kind of impact does the charging effect do to the SEM image? What measures should be taken to overcome this problem? (15)
- (b) How do we get an image by scanning electron microscope (SEM)? Explain with diagram. (12)
- (c) Mention the effect of accelerate voltage for SEM. (08)
8. (a) Explain the working principle of FTIR with suitable diagram. (12)
- (b) What types of vibration modes occurs for H₂O. Draw the diagram of vibration mode. (08)
- (c) For Aluminium, $a = 4.06 \text{ \AA}$, identify the plane for the four peaks and also identify the crystal structure of Aluminium. X ray diffraction was carried out using Cu- K_{α} (1.54 Å) radiation. The XRD peaks are given in figure 8(c). (15)

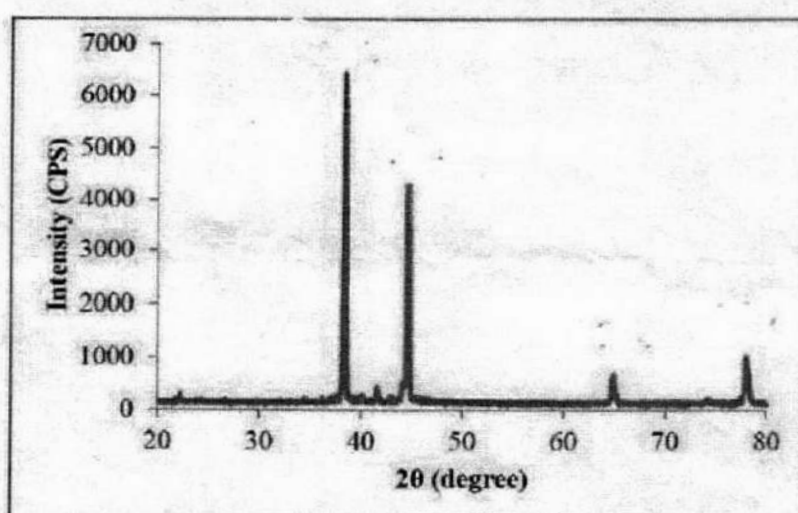


Figure 8(c)

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

MSE 3203
(Ceramics and Glass Engineering)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) State Pauling's rule 2 and explain how this rule can be applied for SiO_2 . (10)
(b) "Point defects are responsible for the alteration of the mechanical properties of ceramics"- Explain in brief. (12)
(c) Estimate the number of Schottky defects in MgO (rocksalt structure) at $600^\circ C$. The enthalpy of formation of the defect is 100 kJ/mole, and the entropy of formation is $6.6R$. The density and molecular weight are 3.58 g/cm^3 and 40.3 g/mole , respectively. (Consider $R=k$). (13)
2. (a) Write down the following defect equations: (15)
(i) Schottky defect for Mn_3O_4 .
(ii) ZnO incorporated into Al_2O_3 .
(iii) Incorporating KCl in $CaCl_2$.
(b) "Most single crystals and all polycrystalline ceramics fracture in a brittle mode with no plastic deformation"-Justify the statement. (10)
(c) How can you reduce the surface flaws in ceramic materials? (10)
3. (a) Why are ceramic materials susceptible to catastrophic failure? (05)
(b) What are the process of toughening mechanism for ceramics? Describe the mechanism of crack shielding and crack bridging methods? (15)
(c) How does Si_3N_4 work as a self-reinforcing material for ceramic toughening? Explain with appropriate figure. (15)
4. (a) Describe the mechanism of creep in ceramics. (10)
(b) Why is designing with brittle ceramics difficult than designing with metals? Justify using strength distribution curve. (10)
(c) A set of ceramic bars of 5mm cross-section and 100mm length are tested in 3-point bend with the outer loading points spanning 90mm. This data was analyzed to give a mean MOR of 380MPa and a Weibull modulus of 14. (15)

Determine the mean expected stress for an identical sample set tested in 4-point bend with the outer and inner loading points spanning 90mm and 30mm respectively.

Section B

(Answer ANY THREE questions from this section in **answer script B**)

5. (a) Write down one example of spinel ferrite, inverse spinel ferrite, and mixed spinel ferrite. (14)
Explain ferrimagnetism phenomena in a spinel ferrite (Fe_3O_4).
(b) Calculate the magnetic moment in Bohr magneton (n_B) and saturation magnetization, (11)
 $M_s(\text{emu/gm})$ of $Ni_{0.75}Zn_{0.25}Fe_2O_4$.
(c) Differentiate between soft ferrite and hard ferrite ceramic materials. (10)
6. (a) Explain the effect of temperature on carrier concentration and conductivity of the extrinsic (10)
ceramic semiconductor.
(b) “ZnO is one of the latest ceramics under research for semiconducting applications”-Explain (15)
the statement.
(c) Why do the most of the materials show superconductivity at low temperatures? Explain with (10)
BCS theory.
7. (a) Describe the polarization versus electric field characteristics of non-linear dielectric ceramics (12)
with an appropriate figure.
(b) For the synthesis of substances that are low in melting point and high in vapor pressure which (13)
chemical synthesis process of ceramic powder will you prefer? Briefly describe the steps of
this synthesis process.
(c) Which ceramic fabrication technique will you prefer for the production of thin sheets of a (10)
flexible tape? Briefly describe this process.
8. (a) What is meant by safety glass? Write down the chemical composition and application of (10)
safety glass.
(b) What is float glass? Briefly describe the production of float glass and its applications. (15)
(c) What type of glasses are used for chemical industries and why? (04)
(d) Suppose that you work for a chemical industries and you are in need to increase the chemical (06)
durability of glass, how you do that?

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MSE 3205
(Corrosion and Surface Engineering)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
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Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Why indirect loss is difficult to assess? Explain in brief. (10)
(b) Compare and contrast concentration cells and differential temperature cells. (10)
(c) Calculate the emf of the cell $Sn/Sn^{2+} // Ti^{2+}/Ti$; at 25⁰ C and also show the polarity of the cell from the free energy change with the help of schematic diagram. The standard potential of $Sn^{2+} + 2e^- \rightarrow Sn$ and $Ti^{2+} + 2e^- \rightarrow Ti$ are -0.136 V and -1.630 V respectively. Given $R=8.314 J/(deg\ mol)$ and $F=96500\ C/eq$. (15)
2. (a) "Li has virtually no resistance to oxidation but Li improves the oxidation resistance of Ni when added in small amounts"-Justify with necessary diagrams. (15)
(b) For each of the metal listed below, compute the Pilling-Bedworth ratio. Which oxide scale that forms on the surface is the most protective and which one is the least protective and why? (10)

Metal	Atomic Wt.	Metal Density (g/cc)	Metal oxide	Oxide density (g/cc)
Zr	90.6	6.51	ZrO ₂	5.89
Bi	208	9.80	Bi ₂ O ₃	8.90
Al	27	2.67	Al ₂ O ₃	3.90

- (c) Explain how metallurgical variables affect the pitting corrosion and what measures can be taken for the prevention of pitting damage. (10)
3. (a) What is filiform corrosion? Discuss the mechanism of filiform corrosion. How can we prevent it? (16)
(b) Describe with necessary figures, how pit depth and specimen area would affect the evaluation of pitting damage. (13)
(c) Filiform corrosion does not destroy the metallic component, however it is a major problem in the canning industry. Explain the reason. (06)
4. (a) As a materials engineer, will you suggest using clean steel to fight against hydrogen embrittlement? Justify your suggestion. (06)
(b) Suggest a cast iron type that can be used in line pipe for carrying water. Justify your choice. (05)
(c) Amount of oxidizer concentration required to cause passivation is greater than that required to maintain passivity. Justify the assertion with necessary figures. (17)
(d) What measures can be taken to prevent hydrogen blistering? (07)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Electroless plating is necessary for deposition on a polymer substrate. Justify with diagram. (10)
(b) Electroplating can improve the corrosion resistance of steel. Explain it with necessary diagram. (15)
(c) Differentiate between Galvanizing process and Tinning process. (05)
(d) How can we improve fatigue wear resistance? (05)
6. (a) Compare and construct physical vapour deposition with chemical vapour deposition. (12)
(b) Mention which coatings are applied on the following applications: (i) food cans (ii) electrical contacts for switch (iii) cutlery. Give reasons to your answer. (15)
(c) Why is surface modification of materials essential? (08)
7. (a) In plating a continuous strip of 1 m long and 0.9 m wide with Sn from an acid solution (valence charge 2), the desired strip speed is 500 m/min, current efficiency is 100% and the thickness of Sn is to be 0.4 μm (each side) at a current density of 5000 A/m². How long must the plating tank be, assuming that the strip passes straight through the tank with no return bends? ($\rho_{\text{Sn}}=7.31 \text{ g/cc}$, $A_{\text{Sn}}=118.7$). (10)
(b) To deposit a high melting point metal such as Cr, which PVD method will you choose? (15)
Describe the method you have chosen.
(c) Describe the factors affecting electroplating. (10)
8. (a) State different tests to judge the quality of coating. (10)
(b) Explain the coefficient of friction as a function of temperature for cobalt. (10)
(c) Derive equation expressing the ploughing component of the coefficient of friction for a cylindrical shaped wear particle in contact with a softer body (consider both upright position and transverse position). (15)

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MSE 3209
(Extractive Metallurgy)

Time: 3 hours

Full Marks: 210

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Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

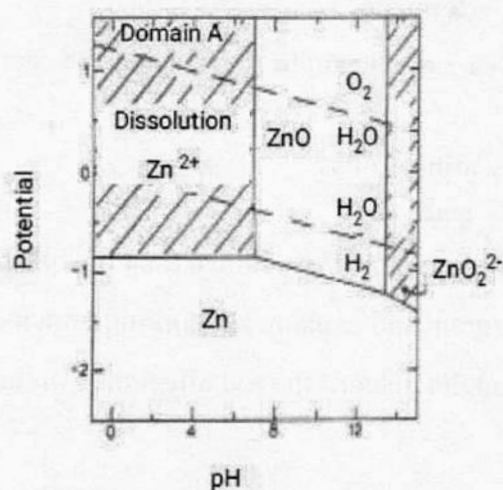
1. (a) "All ores are mineral but all minerals are not ore"-Justify this statement. (07)
(b) Classify the ores. What is meant by run of ore and reduction ratio? How does reduction ratio relate with crushing performance? (13)
(c) Fine particle size gives more accurate results than the coarse size in ore sampling process-Justify. (10)
(d) What do you mean by grindability? (05)
2. (a) What is meant by comminution? Write down the principle of comminution. (10)
(b) Draw a schematic diagram and explain the working principle of the Warman Cychosizer. (15)
(c) How do the length and diameter of the rod affect the grinding performance in a rod mill? (10)
3. (a) Explain the principle of hydrocyclone method with a neat sketch. (15)
(b) Distinguish between free settling and hindered settling. (10)
(c) Which materials are used to fabricate screening surface? Differentiate between bolt in screening and modular screening surface. (10)
4. (a) What is collector in froth flotation? Classify the collectors. (10)
(b) Differentiate between coagulation and flocculation process in sedimentation. (07)
(c) What is the last operation performed in mineral processing plant? (08)
(d) Briefly discuss one of the environmental friendly tailing disposal method. (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) In which conditions will you choose the pyrometallurgy or hydrometallurgy process for metal extractions? Explain in brief. (12)
(b) Describe the principle of metallothermic reduction. (10)
(c) A concentrate has the following composition, % Cu-14; Fe-32; S-38; SiO₂-4; CaO-1; Zn-4; undetermined-6; moisture content-3. A desulphurization of 65 percent is to be achieved. Assume that a small excess of air is used and all of the iron is bound as Fe₃O₄. Determine the composition of the yield. (13)

6. (a) Which hydrometallurgy process will you prefer for the treatment of large quantities of low-grade ore? Describe this process. (13)
- (b) Briefly describe the kinetics of leaching in case of complete dissolution and incomplete dissolution. (13)
- (c) Write down the chemistry of Aluminium electrowinning process. (09)
7. (a) Write down the differences between electrowinning and electrorefining process. (07)
- (b) Draw the potential/ p^H diagram of the Zn-H₂O system shown in figure 7(b) and select a point in this potential/ p^H diagram where the Zn ion can dissolve in water, and write down the corresponding cathodic and anodic reactions. (13)



Potential/pH Diagram for the Zn-H₂O System
(for a metal ion concentration of 10^{-3})

Figure 7(b)

- (c) Mention two names of primary gold sources for gold extraction. Also, briefly describe any gold extraction process from primary sources with a neat sketch. (15)
8. (a) Discuss the factors that cause loss of current efficiency during electrolytes of molten salts. (10)
- (b) Draw and briefly describe series and parallel arrangements of electrodes in the electrolytic cell. (13)
- (c) Calculate the thickness of chromium plating deposited in 2.5 minutes at a current density of 250 mA/cm² and current efficiency of 28%. Density of Cr=7.1 g/cc. (12)

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IEM 3227
(Industrial Management)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) What is meant by Industrial Management? Describe the functions of management. (08)
(b) Write down the differences between delegation and decentralization. (08)
(c) Write down the differences between management and administration. (10)
(d) Define wage, salary and incentive. (09)
2. (a) Describe the contingency theory of Leadership. (12)
(b) Briefly explain Fayol's 14 principles of management. (14)
(c) A fiber optics testing device is to be DDB depreciated. It has a first cost of \$100000 and an estimated salvage of \$15000 after 12 years. (09)
(i) Calculate the depreciation & book value for years 2 & 8.
(ii) Calculate the implied salvage value after 12 years.
3. (a) Describe ERG theory of motivation. (12)
(b) Describe the conditions for effective functioning of committee organization structure. (13)
(c) Define span of management. Elucidate the factors of span of management. (10)
4. (a) Write down the advantages and disadvantages of 'Rowan' wage incentive plan. (10)
(b) Describe the classical approach of organization. (10)
(c) Define project organization. What are the characteristics of project organization? (10)
(d) Briefly describe the four factors of leadership. (05)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

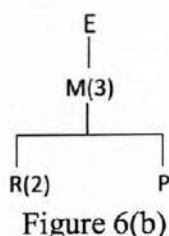
5. (a) Define forecasting. List and explain the types of forecasting in decision making. (10)
(b) Healthy Hamburgers has a chain of 12 stores in northern Illinois. Sales figures and profits for the stores are given in the following table 5(b). Obtain a regression line for the data and predict the profit for a store assuming sales of \$10 million. Both sales and profits are in millions. (15)

Unit Sales	7	2	6	4	14	15	16	12	14	20	15	7
Profits	0.15	0.1	0.13	0.15	0.25	0.27	0.24	0.2	0.27	0.44	0.34	0.17

Table 5(b)

- (c) Define EPQ. How does EOQ differ with EPQ? (10)

6. (a) The maintenance department of a large hospital uses about 816 cases of liquid cleaners annually. (18)
Ordering costs are \$12, carrying costs are \$4 per case a year. Price schedule indicates that orders of less than 50 cases will cost \$20 per case, 50 to 79 cases will cost \$18 per case, 80 to 99 cases will cost \$17 per case and larger orders will cost \$16 per case. Determine the optimal order quantity and total cost.
- (b) The product structure tree of end item E is given in the following figure 6(b). The manager (17)
wants to know the material requirements for ordered part R that will be needed to complete 120 units of E by the start of week 5. Lead times for E is one week, for M is one week and for R and P is two weeks. There is a scheduled receipt of 60 units of M at the end of week 1 and 100 units of R at the start of week 1. Lot for lot ordering is used.

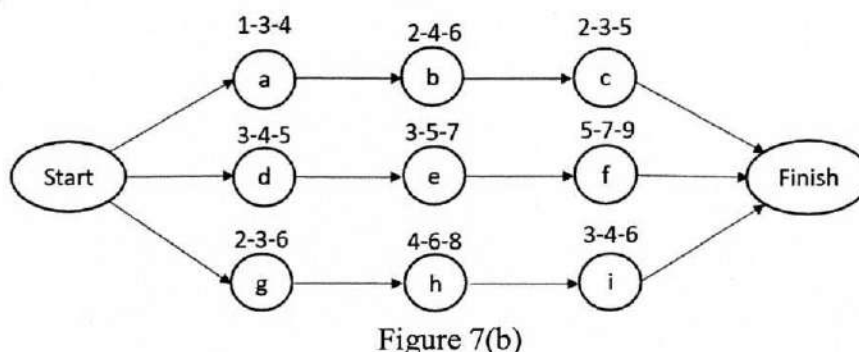


7. (a) The following table contains information on the cost to run three jobs on four available (17)
machines. Determine an assignment plan that will minimize costs and compute that minimized cost.

Machine \ Job	A	B	C	D
1	12	16	14	10
2	9	8	13	7
3	15	12	9	11

Table 7(a)

- (b) The network diagram for a project is shown in the following figure 7(b), with three time (18)
estimates for each activity. Activity times are in weeks. Do the following:
- Compute the expected time for each activity and the expected duration for each path.
 - Identify the critical path.
 - Compute the variance of each activity and the variance and standard deviation of each path.



8. (a) Define MIS. Differentiate between data and information. (10)
- (b) Briefly explain about Retrieval CAPP system with necessary figure. (15)
- (c) Write down the major differences between preventive and breakdown maintenance. (10)